OBC v1.2 Datasheet

High-level overview

- Features an STM32G431RBT6 MCU
  - ARM Cortex-M4
  - 128 kBytes of Flash
  - LQFP-64 Package
- HSE: 24MHz. LSE: 32.768kHz
-

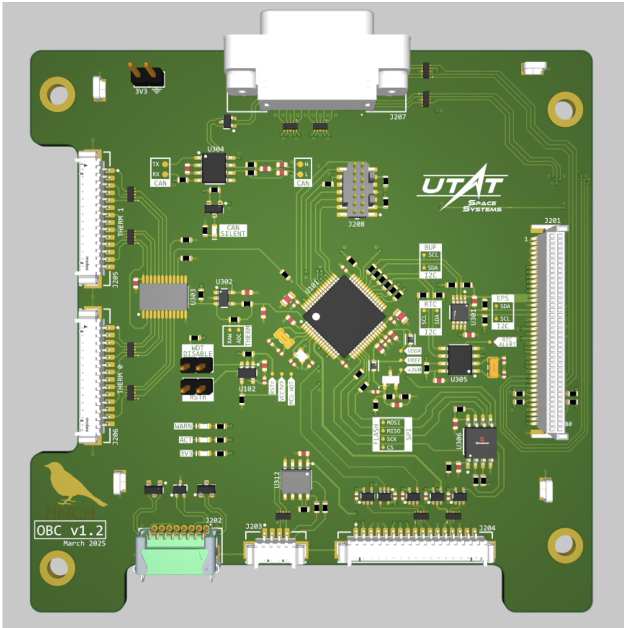

Figure 1: OBC v1.2

Introduction

The **datasheet** document class makes it easy to write great looking data sheets using the LaTeX typesetting system. It follows the classic style used by most manufacturers of electronic components.

You can download the document class from latex-datasheet-template GitHub repository. The repository includes this example datasheet as **example.tex** and the document class as **datasheet.cls**. You can build the PDF document using command `latexmk -pdf`.

Device Overview

Below is the block diagram that represents th high-level specification of the On-Board Computer.

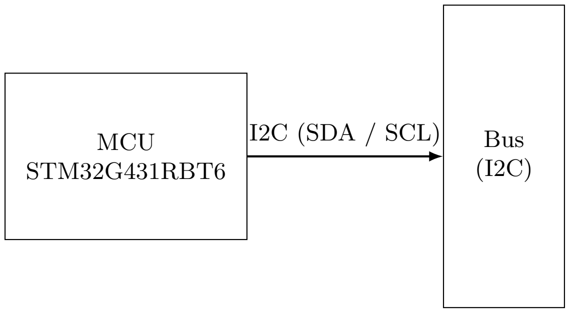

Figure 2: Block diagram: MCU and I2C bus

Specifications

TODO: cover electrical specifications such as voltage ratings and power requirements

Interconnects and Pinout

TODO: intro

Bus

The table below maps the J201 connector pins found in the Protel netlist to their net labels.

Pin	Net
1	NetJ201_1
2	NetJ201_1
3	NetJ201_3
4	NetJ201_3
5	NetJ201_5
6	NetJ201_5
7	SRSVBUS
8	SRSVBUS
9	ADCSVCC5V2
10	ADCSVCC5V2
11	ADCSVCC3V3
12	ADCSVCC3V3
13	VCC3V3
14	VCC3V3
15	GND
16	EPS_I2C_B.SDA

Pin	Net
17	EPS_I2C_B.SCL
18	GND
19	EPS_GPIO0
20	EPS_GPIO2
21	EPS_GPIO4
22	EPS_GPIO6
23	GND
24	RF_RS422.PAY_RX_P
25	RF_RS422.PAY_RX_N
26	GND
27	STAR_RS485.STAR_RX_P
28	STAR_RS485.STAR_RX_N
29	GND
30	PAY_DEBUG.SWCLK
31	PAY_DEBUG.SWDIO
32	PAY_DEBUG.SWO
33	GND
34	BUS_GPIO0
35	BUS_GPIO2
36	BAT_SEL
37	GND
38	GSEVCC
39	GSEVCC
40	GSEVCC
41	GND
42	GND
43	GND
44	GND
45	GND
46	GND
47	GND
48	GND
49	GND
50	GND
51	GND
52	GND
53	GND
54	GND
55	GND

Pin	Net
56	CANBUS.CAN_H
57	CANBUS.CAN_L
58	GND
59	EPS_GPIO1
60	EPS_GPIO3
61	EPS_GPIO5
62	EPS_GPIO7
63	GND
64	RF_RS422.PAY_TX_N
65	RF_RS422.PAY_TX_P
66	GND
67	STAR_RS485.STAR_TX_N
68	STAR_RS485.STAR_TX_P
69	GND
70	PAY_DEBUG.UART_GSE_RX
71	PAY_DEBUG.UART_GSE_TX
72	PAY_DEBUG.RESET_n
73	GND
74	BUS_GPIO1
75	BUS_GPIO3
76	GNSS_PPS_B
77	GND
78	GND
79	GND
80	GND
MP1	GND
MP2	GND

Power

EPS connection

GPIO

GNSS

CANbus

RS-422

RS-485

PAY Debug

ADCS

TODO: specify JTAG pinout and reference layout position; add the image of the connector and the BOM part. Make sure to cover the protection circuitry in the corresponding subsection.

JTAG

TODO: specify JTAG pinout and reference layout position; add the image of the connector and the BOM part

SRS-4

TODO: specify the pinout and reference layout position; add the image of the connector and BOM part

Thermistors

TODO: specify the pinout and reference layout position; add the image of the connector and BOM part

TODO: describe thermistor muxes used by the OBC, use case, part numbers, and the overall design with the opamps

Umbillical

TODO: specify the pinout and reference layout position.

Functional Description

MCU

TODO: describe some high level details about the MCU selected for the board

External Clocks

TODO: describe the external clocks and the reason for use, specify how they are connected to the MCU

External Flash Memory

TODO: descrive the external clocks and the reason for use, specify how its connected to the MCU.

RTC

TODO: describe the RTC usecase and specify how it is connected to the MCU

Real time clock (RTC) is a battery-powered external clock available to OBC for timekeeping and synchronization purposes.

Mechanical and Layout

TODO: add a isometric view of the board with max height, dimentions and drill holes.

TODO: add description of the polygon pours used for power dissipation

Manifacturing

Bill of Materials

Test Points

TODO: describe available test points within the design and cross reference corresponding subsections.

Bringup Log

TODO: specify tests that have been conducted (reference to notion)